14. First let $u = e^{-\theta}$, $dv = \cos 2\theta\, d\theta \quad \Rightarrow \quad du = -e^{-\theta}\, d\theta$, $v = \frac{1}{2}\sin 2\theta$. Then

$I = \int e^{-\theta}\cos 2\theta\, d\theta = \frac{1}{2}e^{-\theta}\sin 2\theta - \int \frac{1}{2}\sin 2\theta\left(-e^{-\theta}\, d\theta\right) = \frac{1}{2}e^{-\theta}\sin 2\theta + \frac{1}{2}\int e^{-\theta}\sin 2\theta\, d\theta.$

Next let $U = e^{-\theta}$, $dV = \sin 2\theta\, d\theta \quad \Rightarrow \quad dU = -e^{-\theta}\, d\theta$, $V = -\frac{1}{2}\cos 2\theta$, so

$\int e^{-\theta}\sin 2\theta\, d\theta = -\frac{1}{2}e^{-\theta}\cos 2\theta - \int\left(-\frac{1}{2}\right)\cos 2\theta\left(-e^{-\theta}\, d\theta\right) = -\frac{1}{2}e^{-\theta}\cos 2\theta - \frac{1}{2}\int e^{-\theta}\cos 2\theta\, d\theta.$

So $I = \frac{1}{2}e^{-\theta}\sin 2\theta + \frac{1}{2}\left[\left(-\frac{1}{2}e^{-\theta}\cos 2\theta\right) - \frac{1}{2}I\right] = \frac{1}{2}e^{-\theta}\sin 2\theta - \frac{1}{4}e^{-\theta}\cos 2\theta - \frac{1}{4}I \quad \Rightarrow$

$\frac{5}{4}I = \frac{1}{2}e^{-\theta}\sin 2\theta - \frac{1}{4}e^{-\theta}\cos 2\theta + C_1 \quad \Rightarrow \quad I = \frac{4}{5}\left(\frac{1}{2}e^{-\theta}\sin 2\theta - \frac{1}{4}e^{-\theta}\cos 2\theta + C_1\right) = \frac{2}{5}e^{-\theta}\sin 2\theta - \frac{1}{5}e^{-\theta}\cos 2\theta + C.$

18. Let $u = \ln t$, $dv = \sqrt{t}\, dt \quad \Rightarrow \quad du = dt/t$, $v = \frac{2}{3}t^{3/2}$. By (6),

$\int_1^4 \sqrt{t}\ln t\, dt = \left[\frac{2}{3}t^{3/2}\ln t\right]_1^4 - \frac{2}{3}\int_1^4 \sqrt{t}\, dt = \frac{2}{3}\cdot 8\cdot \ln 4 - 0 - \left[\frac{2}{3}\cdot\frac{2}{3}t^{3/2}\right]_1^4 = \frac{16}{3}\ln 4 - \frac{4}{9}(8-1) = \frac{16}{3}\ln 4 - \frac{28}{9}.$

20. Let $u = \arctan(1/x)$, $dv = dx \quad \Rightarrow \quad du = \dfrac{1}{1+(1/x)^2}\cdot\dfrac{-1}{x^2}\, dx = \dfrac{-dx}{x^2+1}$, $v = x$. Then

$$\int_1^{\sqrt{3}} \arctan(1/x)\, dx = \left[x\arctan\left(\frac{1}{x}\right)\right]_1^{\sqrt{3}} + \int_1^{\sqrt{3}} \frac{x\, dx}{x^2+1} = \sqrt{3}\,\frac{\pi}{6} - 1\cdot\frac{\pi}{4} + \frac{1}{2}\left[\ln(x^2+1)\right]_1^{\sqrt{3}}$$

$$= \frac{\pi\sqrt{3}}{6} - \frac{\pi}{4} + \frac{1}{2}(\ln 4 - \ln 2) = \frac{\pi\sqrt{3}}{6} - \frac{\pi}{4} + \frac{1}{2}\ln\frac{4}{2} = \frac{\pi\sqrt{3}}{6} - \frac{\pi}{4} + \frac{1}{2}\ln 2$$

3. $\int_{\pi/2}^{3\pi/4} \sin^5 x\, \cos^3 x\, dx = \int_{\pi/2}^{3\pi/4} \sin^5 x\, \cos^2 x\, \cos x\, dx = \int_{\pi/2}^{3\pi/4} \sin^5 x\,(1-\sin^2 x)\cos x\, dx \overset{s}{=} \int_1^{\sqrt{2}/2} u^5(1-u^2)\, du$

$= \int_1^{\sqrt{2}/2}(u^5 - u^7)\, du = \left[\frac{1}{6}u^6 - \frac{1}{8}u^8\right]_1^{\sqrt{2}/2} = \left(\frac{1/8}{6} - \frac{1/16}{8}\right) - \left(\frac{1}{6} - \frac{1}{8}\right) = -\frac{11}{384}$

14. $\int_0^{\pi/2} \sin^2 x\, \cos^2 x\, dx = \int_0^{\pi/2} \frac{1}{4}(4\sin^2 x\, \cos^2 x)\, dx = \int_0^{\pi/2} \frac{1}{4}(2\sin x\, \cos x)^2\, dx = \frac{1}{4}\int_0^{\pi/2} \sin^2 2x\, dx$

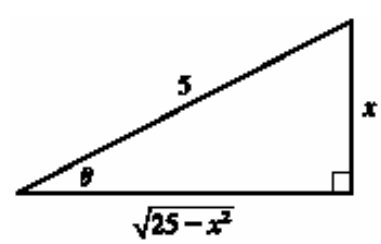
$= \frac{1}{4}\int_0^{\pi/2} \frac{1}{2}(1-\cos 4x)\, dx = \frac{1}{8}\int_0^{\pi/2}(1-\cos 4x)\, dx = \frac{1}{8}\left[x - \frac{1}{4}\sin 4x\right]_0^{\pi/2} = \frac{1}{8}\left(\frac{\pi}{2}\right) = \frac{\pi}{16}$

19. $\int \tan^2 x\, dx = \int(\sec^2 x - 1)\, dx = \tan x - x + C$

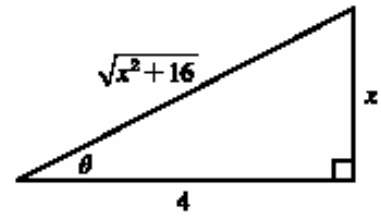
43. Let $x = 5\sin\theta$, so $dx = 5\cos\theta\, d\theta$. Then

$$\int \frac{1}{x^2\sqrt{25-x^2}}\, dx = \int \frac{1}{5^2\sin^2\theta\cdot 5\cos\theta}\, 5\cos\theta\, d\theta = \frac{1}{25}\int \csc^2\theta\, d\theta$$

$$= -\frac{1}{25}\cot\theta + C = -\frac{1}{25}\frac{\sqrt{25-x^2}}{x} + C$$



45. Let $x = 4\tan\theta$, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Then $dx = 4\sec^2\theta\,d\theta$ and



$$\sqrt{x^2+16} = \sqrt{16\tan^2\theta + 16} = \sqrt{16(\tan^2\theta+1)} = \sqrt{16\sec^2\theta}$$
$$= 4\,|\sec\theta| = 4\sec\theta \quad \text{for the relevant values of } \theta.$$

$$\int \frac{dx}{\sqrt{x^2+16}} = \int \frac{4\sec^2\theta\,d\theta}{4\sec\theta} = \int \sec\theta\,d\theta = \ln|\sec\theta + \tan\theta| + C_1 = \ln\left|\frac{\sqrt{x^2+16}}{4} + \frac{x}{4}\right| + C_1$$
$$= \ln\left|\sqrt{x^2+16} + x\right| - \ln|4| + C_1 = \ln\left(\sqrt{x^2+16}+x\right) + C, \text{ where } C = C_1 - \ln 4.$$

(Since $\sqrt{x^2+16} + x > 0$, we don't need the absolute value.)

58. Let $u = \sin t$, $du = \cos t\,dt$. Then

$$\int_0^{\pi/2} \frac{\cos t}{\sqrt{1+\sin^2 t}}\,dt = \int_0^1 \frac{1}{\sqrt{1+u^2}}\,du = \int_0^{\pi/4} \frac{1}{\sec\theta}\sec^2\theta\,d\theta \qquad \left[\begin{array}{c}\text{where } u = \tan\theta,\ du = \sec^2\theta\,d\theta, \\ \text{and } \sqrt{1+u^2} = \sec\theta\end{array}\right]$$
$$= \int_0^{\pi/4} \sec\theta\,d\theta = \Big[\ln|\sec\theta + \tan\theta|\Big]_0^{\pi/4} \qquad [\text{by } (1)]$$
$$= \ln\left(\sqrt{2}+1\right) - \ln(1+0) = \ln\left(\sqrt{2}+1\right)$$

12. $\dfrac{x-1}{x^2+3x+2} = \dfrac{A}{x+1} + \dfrac{B}{x+2}$. Multiply both sides by $(x+1)(x+2)$ to get $x - 1 = A(x+2) + B(x+1)$. Substituting -2 for x gives $-3 = -B \quad \Leftrightarrow \quad B = 3$. Substituting -1 for x gives $-2 = A$. Thus,

$$\int_0^1 \frac{x-1}{x^2+3x+2}\,dx = \int_0^1 \left(\frac{-2}{x+1} + \frac{3}{x+2}\right)dx = \Big[-2\ln|x+1| + 3\ln|x+2|\Big]_0^1$$
$$= (-2\ln 2 + 3\ln 3) - (-2\ln 1 + 3\ln 2) = 3\ln 3 - 5\ln 2 \quad \left[\text{or } \ln\tfrac{27}{32}\right]$$

21. $\dfrac{5x^2+3x-2}{x^3+2x^2} = \dfrac{5x^2+3x-2}{x^2(x+2)} = \dfrac{A}{x} + \dfrac{B}{x^2} + \dfrac{C}{x+2}$. Multiply by $x^2(x+2)$ to get

$5x^2 + 3x - 2 = Ax(x+2) + B(x+2) + Cx^2$. Set $x = -2$ to get $C = 3$, and take $x = 0$ to get $B = -1$.

Equating the coefficients of x^2 gives $5 = A + C \quad \Rightarrow \quad A = 2$.

So $\displaystyle\int \frac{5x^2+3x-2}{x^3+2x^2}\,dx = \int \left(\frac{2}{x} - \frac{1}{x^2} + \frac{3}{x+2}\right)dx = 2\ln|x| + \frac{1}{x} + 3\ln|x+2| + C.$

24. $\dfrac{x^2-2x-1}{(x-1)^2(x^2+1)} = \dfrac{A}{x-1} + \dfrac{B}{(x-1)^2} + \dfrac{Cx+D}{x^2+1} \quad \Rightarrow$

$x^2-2x-1 = A(x-1)(x^2+1) + B(x^2+1) + (Cx+D)(x-1)^2$. Setting $x=1$ gives $B=-1$. Equating the coefficients of x^3 gives $A=-C$. Equating the constant terms gives $-1=-A-1+D$, so $D=A$, and setting $x=2$ gives $-1=5A-5-2A+A$ or $A=1$. We have

$$\int \frac{x^2-2x-1}{(x-1)^2(x^2+1)}\,dx = \int \left[\frac{1}{x-1} - \frac{1}{(x-1)^2} - \frac{x-1}{x^2+1}\right] dx = \ln|x-1| + \frac{1}{x-1} - \tfrac{1}{2}\ln(x^2+1) + \tan^{-1}x + C.$$

15. $f(y) = \dfrac{1}{1+y^5}$, $\Delta y = \dfrac{3-0}{6} = \dfrac{1}{2}$

(a) $T_6 = \frac{1}{2\cdot 2}\left[f(0) + 2f\left(\frac{1}{2}\right) + 2f\left(\frac{2}{2}\right) + 2f\left(\frac{3}{2}\right) + 2f\left(\frac{4}{2}\right) + 2f\left(\frac{5}{2}\right) + f(3)\right] \approx 1.064275$

(b) $M_6 = \frac{1}{2}\left[f\left(\frac{1}{4}\right) + f\left(\frac{3}{4}\right) + f\left(\frac{5}{4}\right) + f\left(\frac{7}{4}\right) + f\left(\frac{9}{4}\right) + f\left(\frac{11}{4}\right)\right] \approx 1.067416$

(c) $S_6 = \frac{1}{2\cdot 3}\left[f(0) + 4f\left(\frac{1}{2}\right) + 2f\left(\frac{2}{2}\right) + 4f\left(\frac{3}{2}\right) + 2f\left(\frac{4}{2}\right) + 4f\left(\frac{5}{2}\right) + f(3)\right] \approx 1.074915$

15. $$\int_0^\infty se^{-5s}\,ds = \lim_{t\to\infty}\int_0^t se^{-5s}\,ds = \lim_{t\to\infty}\left[-\frac{1}{5}se^{-5s} - \frac{1}{25}e^{-5s}\right]_0^t \qquad \left[\begin{array}{l}\text{by integration by}\\ \text{parts with } u=s\end{array}\right]$$

$$= \lim_{t\to\infty}\left(-\tfrac{1}{5}te^{-5t} - \tfrac{1}{25}e^{-5t} + \tfrac{1}{25}\right) = 0 - 0 + \tfrac{1}{25} \qquad \text{[by l'Hospital's Rule]}$$

$$= \tfrac{1}{25} \qquad \text{Convergent}$$

32. Integrate by parts with $u=\ln x$, $dv = dx/\sqrt{x} \quad\Rightarrow\quad du = dx/x$, $v = 2\sqrt{x}$.

$$\int_0^1 \frac{\ln x}{\sqrt{x}}\,dx = \lim_{t\to 0^+}\int_t^1 \frac{\ln x}{\sqrt{x}}\,dx = \lim_{t\to 0^+}\left(\left[2\sqrt{x}\ln x\right]_t^1 - 2\int_t^1 \frac{dx}{\sqrt{x}}\right) = \lim_{t\to 0^+}\left(-2\sqrt{t}\ln t - 4\left[\sqrt{x}\right]_t^1\right)$$

$$= \lim_{t\to 0^+}\left(-2\sqrt{t}\ln t - 4 + 4\sqrt{t}\right) = -4$$

since $\displaystyle\lim_{t\to 0^+}\sqrt{t}\ln t = \lim_{t\to 0^+}\frac{\ln t}{t^{-1/2}} \overset{\text{H}}{=} \lim_{t\to 0^+}\frac{1/t}{-t^{-3/2}/2} = \lim_{t\to 0^+}\left(-2\sqrt{t}\right) = 0.$ Convergent

42. For $x \ge 1$, $\dfrac{2+e^{-x}}{x} > \dfrac{2}{x}$ [since $e^{-x} > 0$] $> \dfrac{1}{x}$. $\displaystyle\int_1^\infty \frac{1}{x}\,dx$ is divergent by (2) with $p = 1 \le 1$, so $\displaystyle\int_1^\infty \frac{2+e^{-x}}{x}\,dx$ is divergent by the Comparison Theorem.

45. $\dfrac{1}{x\sin x} \geq \dfrac{1}{x}$ on $\left(0, \frac{\pi}{2}\right]$ since $0 \leq \sin x \leq 1$. $\displaystyle\int_0^{\pi/2} \frac{dx}{x} = \lim_{t\to 0^+} \int_t^{\pi/2} \frac{dx}{x} = \lim_{t\to 0^+} \left[\ln x\right]_t^{\pi/2}$.

But $\ln t \to -\infty$ as $t \to 0^+$, so $\displaystyle\int_0^{\pi/2} \frac{dx}{x}$ is divergent, and by the Comparison Theorem, $\displaystyle\int_0^{\pi/2} \frac{dx}{x\sin x}$ is also divergent.

47. $\displaystyle\int_0^\infty \frac{dx}{\sqrt{x}\,(1+x)} = \int_0^1 \frac{dx}{\sqrt{x}\,(1+x)} + \int_1^\infty \frac{dx}{\sqrt{x}\,(1+x)} = \lim_{t\to 0^+} \int_t^1 \frac{dx}{\sqrt{x}\,(1+x)} + \lim_{t\to\infty} \int_1^t \frac{dx}{\sqrt{x}\,(1+x)}$.

Now $\displaystyle\int \frac{dx}{\sqrt{x}\,(1+x)} = \int \frac{2u\,du}{u(1+u^2)}$ $\quad [u = \sqrt{x}, x = u^2, dx = 2u\,du]$

$$= 2\int \frac{du}{1+u^2} = 2\tan^{-1} u + C = 2\tan^{-1}\sqrt{x} + C,$$

so $\displaystyle\int_0^\infty \frac{dx}{\sqrt{x}\,(1+x)} = \lim_{t\to 0^+} \left[2\tan^{-1}\sqrt{x}\right]_t^1 + \lim_{t\to\infty} \left[2\tan^{-1}\sqrt{x}\right]_1^t$

$$= \lim_{t\to 0^+} \left[2\left(\tfrac{\pi}{4}\right) - 2\tan^{-1}\sqrt{t}\right] + \lim_{t\to\infty} \left[2\tan^{-1}\sqrt{t} - 2\left(\tfrac{\pi}{4}\right)\right] = \tfrac{\pi}{2} - 0 + 2\left(\tfrac{\pi}{2}\right) - \tfrac{\pi}{2} = \pi.$$

4. $y^2 = 4(x+4)^3,\ \ y > 0 \ \Rightarrow\ y = 2(x+4)^{3/2} \ \Rightarrow\ dy/dx = 3(x+4)^{1/2} \ \Rightarrow$

$1 + (dy/dx)^2 = 1 + 9(x+4) = 9x + 37$. So

$$L = \int_0^2 \sqrt{9x+37}\,dx \quad \begin{bmatrix} u = 9x+37, \\ du = 9\,dx \end{bmatrix} \quad = \int_{37}^{55} u^{1/2}\left(\frac{1}{9}\,du\right) = \tfrac{1}{9}\cdot\tfrac{2}{3}\left[u^{3/2}\right]_{37}^{55} = \tfrac{2}{27}\left(55\sqrt{55} - 37\sqrt{37}\right)$$

5. $y = \dfrac{x^5}{6} + \dfrac{1}{10x^3} \ \Rightarrow\ \dfrac{dy}{dx} = \dfrac{5}{6}x^4 - \dfrac{3}{10}x^{-4} \ \Rightarrow$

$1 + (dy/dx)^2 = 1 + \frac{25}{36}x^8 - \frac{1}{2} + \frac{9}{100}x^{-8} = \frac{25}{36}x^8 + \frac{1}{2} + \frac{9}{100}x^{-8} = \left(\frac{5}{6}x^4 + \frac{3}{10}x^{-4}\right)^2$. So

$$L = \int_1^2 \sqrt{\left(\tfrac{5}{6}x^4 + \tfrac{3}{10}x^{-4}\right)^2}\,dx = \int_1^2 \left(\tfrac{5}{6}x^4 + \tfrac{3}{10}x^{-4}\right)dx = \left[\tfrac{1}{6}x^5 - \tfrac{1}{10}x^{-3}\right]_1^2$$
$$= \left(\tfrac{32}{6} - \tfrac{1}{80}\right) - \left(\tfrac{1}{6} - \tfrac{1}{10}\right) = \tfrac{31}{6} + \tfrac{7}{80} = \tfrac{1261}{240}$$

7. $x = \frac{1}{3}\sqrt{y}\,(y-3) = \frac{1}{3}y^{3/2} - y^{1/2} \ \Rightarrow\ dx/dy = \frac{1}{2}y^{1/2} - \frac{1}{2}y^{-1/2} \ \Rightarrow$

$1 + (dx/dy)^2 = 1 + \frac{1}{4}y - \frac{1}{2} + \frac{1}{4}y^{-1} = \frac{1}{4}y + \frac{1}{2} + \frac{1}{4}y^{-1} = \left(\frac{1}{2}y^{1/2} + \frac{1}{2}y^{-1/2}\right)^2$. So

$$L = \int_1^9 \left(\tfrac{1}{2}y^{1/2} + \tfrac{1}{2}y^{-1/2}\right)dy = \tfrac{1}{2}\left[\tfrac{2}{3}y^{3/2} + 2y^{1/2}\right]_1^9 = \tfrac{1}{2}\left[\left(\tfrac{2}{3}\cdot 27 + 2\cdot 3\right) - \left(\tfrac{2}{3}\cdot 1 + 2\cdot 1\right)\right]$$
$$= \tfrac{1}{2}\left(24 - \tfrac{8}{3}\right) = \tfrac{1}{2}\left(\tfrac{64}{3}\right) = \tfrac{32}{3}$$

27. $y = 2x^{3/2} \quad \Rightarrow \quad y' = 3x^{1/2} \quad \Rightarrow \quad 1 + (y')^2 = 1 + 9x$. The arc length function with starting point $P_0(1, 2)$ is

$s(x) = \int_1^x \sqrt{1+9t}\,dt = \left[\frac{2}{27}(1+9t)^{3/2}\right]_1^x = \frac{2}{27}\left[(1+9x)^{3/2} - 10\sqrt{10}\right]$.

2. $\dfrac{dy}{dx} = \dfrac{\sqrt{x}}{e^y} \quad \Rightarrow \quad e^y\,dy = \sqrt{x}\,dx \quad \Rightarrow \quad \int e^y\,dy = \int x^{1/2}\,dx \quad \Rightarrow \quad e^y = \frac{2}{3}x^{3/2} + C \quad \Rightarrow \quad y = \ln\left(\frac{2}{3}x^{3/2} + C\right)$

7. $\dfrac{du}{dt} = 2 + 2u + t + tu \quad \Rightarrow \quad \dfrac{du}{dt} = (1+u)(2+t) \quad \Rightarrow \quad \displaystyle\int \frac{du}{1+u} = \int (2+t)\,dt \quad [u \neq -1] \quad \Rightarrow$

$\ln|1+u| = \frac{1}{2}t^2 + 2t + C \quad \Rightarrow \quad |1+u| = e^{t^2/2+2t+C} = Ke^{t^2/2+2t}$, where $K = e^C \quad \Rightarrow \quad 1 + u = \pm Ke^{t^2/2+2t} \quad \Rightarrow$

$u = -1 \pm Ke^{t^2/2+2t}$ where $K > 0$. $\quad u = -1$ is also a solution, so $u = -1 + Ae^{t^2/2+2t}$, where A is an arbitrary constant.

11. $x\cos x = (2y + e^{3y})\,y' \quad \Rightarrow \quad x\cos x\,dx = (2y + e^{3y})\,dy \quad \Rightarrow \quad \int(2y + e^{3y})\,dy = \int x\cos x\,dx \quad \Rightarrow$

$y^2 + \frac{1}{3}e^{3y} = x\sin x + \cos x + C$ [where the second integral is evaluated using integration by parts]. Now $y(0) = 0 \quad \Rightarrow$

$0 + \frac{1}{3} = 0 + 1 + C \quad \Rightarrow \quad C = -\frac{2}{3}$. Thus, a solution is $y^2 + \frac{1}{3}e^{3y} = x\sin x + \cos x - \frac{2}{3}$. We cannot solve explicitly for y.

37. The differential equation is a logistic equation with $k = 0.00008$, carrying capacity $M = 1000$, and initial population $y_0 = P(0) = 100$. So Equation 8 gives the population at time t as

$$P(t) = \frac{100 \cdot 1000}{100 + (1000 - 100)e^{-0.08t}} = \frac{100,000}{100 + 900e^{-0.08t}} = \frac{1000}{1 + 9e^{-0.08t}}$$

So the population sizes when $t = 40$ and 80 are

$$P(40) = \frac{1000}{1 + 9e^{-3.2}} \approx 731.6 \qquad P(80) = \frac{1000}{1 + 9e^{-6.4}} \approx 985.3$$

The population reaches 900 when $\dfrac{1000}{1 + 9e^{-0.08t}} = 900$. Solving this equation for t, we get $1 + 9e^{-0.08t} = \frac{10}{9} \quad \Rightarrow$

$e^{-0.08t} = \frac{1}{81} \quad \Rightarrow \quad -0.08t = \ln\frac{1}{81} = -\ln 81 \quad \Rightarrow \quad t = \frac{\ln 81}{0.08} \approx 54.9$. So the population reaches 900 when t is approximately 55.